Proposed Modeling Approach: Behavioral Predictors of Weight Change (NHANES 2021–2023)

Objective and data

We will identify which behavioral characteristics best predict 1-year % weight change among U.S. adults who tried to lose weight, beyond demographics and starting BMI. The sample is 2,794 adults aged >=20 from NHANES August 2021–August 2023 who reported trying to lose weight in the past 12 months and completed the MEC exam (pregnant women excluded). The outcome is self-reported % weight change, (current − 1 year ago) ÷ 1 year ago × 100, restricted to ±50%.

Predictors (27)

Covariates variables (15): Age, sex, race/ethnicity, education, income-to-poverty ratio, marital status, household size, US-born, veteran, BMI 1 year ago.

Behaviors variables (12): Moderate and vigorous activity (min/week), sedentary hours/day, PHQ-9 score, drinks/week, average sleep hours, smoking status, meal-planning and food-shopping roles.

Models

Two model families, each fitted twice, once on covariates only and once on covariates plus behaviors.

- LASSO / RIDGE/ elastic-net regression
- Gradient boosting (XGBoost or LightGBM)